Chapter 2. Topological Spaces and Continuous Functions

Bases, order, product, subspace, metric and quotient topologies; closed sets; continuity

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2.1 Topological spaces and bases

Definition (Topology)

A **topology** on a set X is a collection $\mathcal{T}$ of subsets (the **open sets**) containing $\emptyset$ and X and closed under arbitrary unions and finite intersections.

Examples: the discrete topology (all subsets), the indiscrete topology $\{\emptyset, X\}$, the finite complement topology (U open iff $X - U$ is finite or $U = \emptyset$). $\mathcal{T}'$ is **finer** than $\mathcal{T}$ if $\mathcal{T} \subset \mathcal{T}'$.

Definition (Basis)

A **basis** is a collection $\mathcal{B}$ of subsets such that every x lies in some $B \in \mathcal{B}$, and if $x \in B_1 \cap B_2$ there is $B_3 \in \mathcal{B}$ with $x \in B_3 \subset B_1 \cap B_2$. The topology it generates consists of all unions of elements of $\mathcal{B}$. A **subbasis** is any collection whose union is X; finite intersections of its members form a basis.

Lemma 2.1

(a) If $\mathcal{C}$ is a collection of open sets such that every open U and $x \in U$ admit $C \in \mathcal{C}$ with $x \in C \subset U$, then $\mathcal{C}$ is a basis for the topology. (b) The topology of $\mathcal{B}'$ is finer than that of $\mathcal{B}$ iff for every $B \in \mathcal{B}$ and $x \in B$ there is $B' \in \mathcal{B}'$ with $x \in B' \subset B$.

On $\mathbb{R}$: the **standard topology** has basis the open intervals; the **lower limit topology** $\mathbb{R}_\ell$ has basis $[a, b)$; the **K-topology** $\mathbb{R}_K$ has basis (a, b) and $(a, b) - K$, $K = \{1/n\}$. Both $\mathbb{R}_\ell$ and $\mathbb{R}_K$ are strictly finer than the standard topology and are not comparable with each other.

2.2 Order, product and subspace topologies

For a simply ordered set X with more than one element, the **order topology** has basis the open intervals (a, b) together with $[a_0, b)$ or $(a, b_0]$ if X has a least element a_0 or largest element b_0. The open rays form a subbasis. Examples: $\mathbb{R}$; $\mathbb{Z}_+$ (discrete); the dictionary order on $\mathbb{R} \times \mathbb{R}$; $\{1, 2\} \times \mathbb{Z}_+$ (not discrete: 2×1 has no smallest open neighbourhood avoiding $1 \times \mathbb{Z}_+$).

The **product topology** on $X \times Y$ has basis $U \times V$ with U, V open; if $\mathcal{B}, \mathcal{C}$ are bases, $\{B \times C\}$ is a basis. Equivalently it has subbasis $\pi_1^{-1}(U)$, $\pi_2^{-1}(V)$.

For $Y \subset X$, the **subspace topology** is $\{Y \cap U : U \text{ open in } X\}$. If Y is open in X, open sets of Y are open in X. The product of subspaces is the subspace of the product. But order and subspace topologies can differ: on $Y = [0, 1) \cup \{2\} \subset \mathbb{R}$ the point $\{2\}$ is open in the subspace topology but not in the order topology of Y. They agree when Y is convex (an interval) in X.

2.3 Closed sets and limit points

A is **closed** if $X - A$ is open; closed sets are closed under arbitrary intersections and finite unions. The **closure** $\overline{A}$ is the intersection of all closed sets containing A, the **interior** $\operatorname{Int} A$ the union of all open sets inside A.

Theorem 2.2

$x \in \overline{A}$ iff every open set containing x (equivalently every basis element containing x) meets A. If A' is the set of limit points of A (x such that every neighbourhood of x meets $A - \{x\}$),

then $\overline{A} = A \cup A'$; A is closed iff $A' \subset A$. In a subspace Y, the closure of $A \subset Y$ is $\overline{A} \cap Y$.

Definition (Hausdorff space)

X is **Hausdorff** if distinct points have disjoint neighbourhoods. X satisfies T_1 if finite sets are closed.

In a Hausdorff space finite sets are closed, a sequence converges to at most one point, and in a T_1 space $x \in A'$ iff every neighbourhood of x contains infinitely many points of A. Order topologies, products and subspaces of Hausdorff spaces are Hausdorff. The finite complement topology on $\mathbb{R}$ is T_1 but not Hausdorff, and there the sequence $1, 2, 3, \dots$ converges to every point.

2.4 Continuous functions

Definition (Continuity, homeomorphism)

$f : X \to Y$ is **continuous** if $f^{-1}(V)$ is open for every open V. A continuous bijection with continuous inverse is a **homeomorphism**; an injective continuous map that is a homeomorphism onto its image is an **imbedding**.

Theorem 2.3

The following are equivalent for $f : X \to Y$: (a) f is continuous; (b) $f(\overline{A}) \subset \overline{f(A)}$ for all A; (c) preimages of closed sets are closed; (d) for each x and each neighbourhood V of $f(x)$ there is a neighbourhood U of x with $f(U) \subset V$.

Constructions: constants, inclusions, compositions, restrictions, and maps that are continuous on each set of an open cover are continuous.

Theorem 2.4 (Pasting lemma)

If $X = A \cup B$ with A, B closed and $f : A \to Y$, $g : B \to Y$ are continuous and agree on $A \cap B$, then the combined function is continuous.

Proof. For closed C, $h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$ is a union of two sets closed in A, B, hence in X. ■

$f : A \to X \times Y$ is continuous iff both coordinates are.

2.5 The product topology

For an indexed family $\{X_\alpha\}_{\alpha \in J}$ the **box topology** on $\prod X_\alpha$ has basis $\prod U_\alpha$ with all U_α open, and the **product topology** has basis $\prod U_\alpha$ with $U_\alpha = X_\alpha$ for all but finitely many α (subbasis $\pi_\beta^{-1}(U_\beta)$). They agree for finite products.

Theorem 2.5

With the product topology, $f : A \to \prod X_\alpha$ is continuous iff each $\pi_\alpha \circ f$ is continuous. This fails for the box topology.

Proof. Preimages of subbasis elements are $(\pi_\beta \circ f)^{-1}(U_\beta)$. For the box topology on $\mathbb{R}^\omega$, $f(t) = (t, t, t, \dots)$ has continuous coordinates, but $f^{-1}(\prod(-\frac{1}{n}, \frac{1}{n})) = \{0\}$ is not open. ■

Products and subspaces commute, products of Hausdorff spaces are Hausdorff, and $\prod \overline{A_\alpha} = \overline{\prod A_\alpha}$ in either topology. Unless stated otherwise, products carry the product topology.

2.6 Metric topologies

A metric d gives the topology with basis the ϵ-balls $B_d(x,\epsilon)$; X is **metrizable** if some metric induces its topology. Examples: the Euclidean and square metrics on $\mathbb{R}^n$ give the product topology; the **uniform metric** $\bar{\rho}(x,y) = \sup \min(|x_\alpha - y_\alpha|, 1)$ on $\mathbb{R}^J$ gives a topology between product and box.

Theorem 2.6

$\mathbb{R}^\omega$ with the product topology is metrizable, by $D(x,y) = \sup_n \frac{\min(|x_n - y_n|,1)}{n}$. Countable products of metrizable spaces are metrizable.

Proof. A D-ball of radius ϵ around x contains the product basis set $\prod_{n\le N}(x_n-\epsilon, x_n+\epsilon)\times\mathbb{R}\times\cdots$ with $1/N < \epsilon$, and conversely a product basis set around x restricting finitely many coordinates contains a small D-ball. ■

In metric spaces continuity is the ϵ–δ condition, and the **sequence lemma** holds: $x \in \overline{A}$ iff a sequence in A converges to x (the "if" part in any space; the "only if" needs a countable basis at x). Hence f is continuous iff it preserves convergent sequences. A uniform limit of continuous maps into a metric space is continuous. These facts show that $\mathbb{R}^\omega$ in the box topology and $\mathbb{R}^J$ (J uncountable) in the product topology are not metrizable.

2.7 The quotient topology

A surjective continuous $p : X \to Y$ is a **quotient map** if $U \subset Y$ is open whenever $p^{-1}(U)$ is open. Open or closed continuous surjections are quotient maps. For a partition X^* of X, the **quotient space** carries the finest topology making the projection continuous.

Theorem 2.7

Let $p : X \to Y$ be a quotient map and $g : X \to Z$ constant on each fibre $p^{-1}(y)$. Then $g = f \circ p$ for a unique $f : Y \to Z$, and f is continuous; f is a quotient map iff g is.

Example 2.1

(a) The square $[0,1]^2$ with opposite edges identified is the torus; with all boundary points identified to one point it is the sphere S^2.
(b) Products of quotient maps need not be quotient maps; and quotients of Hausdorff spaces need not be Hausdorff (collapse $\mathbb{Q}$ in $\mathbb{R}$).
(c) $p : \mathbb{R} \to \mathbb{R}/\mathbb{Z} \cong S^1$, $p(t) = (\cos 2\pi t, \sin 2\pi t)$, is a quotient map.